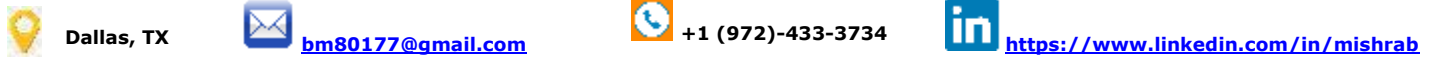


BIBHU MISHRA

DevOps Architect | Cloud Computing, Azure, Kubernetes, Gen AI



DevOps & Cloud Architect with deep expertise in **Azure DevOps, CI/CD automation, Kubernetes (AKS), Terraform, Azure cloud, and application security**. Skilled in designing and implementing **enterprise-scale DevOps solutions** that streamline software delivery, optimize cloud infrastructure, and enhance security and compliance. Experienced in building robust CI/CD pipelines, enabling end-to-end automation, and driving continuous improvement for high-performance software development environments.

SUMMARY

- **DevOps & Cloud Architecture:** Extensive expertise in **Azure DevOps, CI/CD automation, Kubernetes (AKS), Terraform, Azure cloud services**, and application security; skilled in designing scalable, resilient, and secure enterprise systems.
- **Generative AI & LLMops:** Experienced in leveraging **GenAI** to automate **SRE operations**, including retrieval-augmented generation (RAG) and agentic AI chatbots, log analysis, incident classification, and infrastructure optimization. My GenAI POCs are available at [Hugging Face](#).
- **GenAI Tools:** Hands-on with **LangChain, LangGraph, LangSmith, Autogen, MLflow, and ZenML** for intelligent automation, observability, and data-driven system improvement.
- **CI/CD & Infrastructure as Code:** Expertise in **YAML-based CI/CD pipelines, Git branching strategies, Terraform, Ansible**, and automated security scanning for enterprise deployments.
- **Application Development:** Skilled in building **complex enterprise applications and microservices** using **Node.js and Python**, with strong understanding of **microservices architecture and secure coding best practices**.
- **Security & DevSecOps:** Proficient in managing **SAST, DAST, SCA scans** using **SonarQube, CodeQL, Snyk**, enforcing security in repositories, pipelines, and cloud deployments.
- **Monitoring & Observability:** Experienced with **Azure Monitor, Prometheus, Grafana, and Application Insights** to implement proactive monitoring, troubleshooting, and performance optimization.
- **Cloud Platforms & Services:** Deep knowledge of **Azure App Services, Function Apps, Logic Apps, Storage, SQL Servers, Cosmos DB, Event Grid, Service Bus, Azure Front Door, Load Balancers, Azure AD, and networking**.
- **Domain Expertise:** Extensive experience in **Healthcare (EHR, HL7, HIPAA, EMR systems)** and **Content Management (CMS, digital marketing, enterprise content solutions)**, delivering compliant and secure solutions tailored to business needs.
- **Leadership & Mentorship:** Led enterprise DevOps transformations, migrated **40+ enterprise websites**, facilitated GenAI enablement practices, and mentored teams on best practices in cloud, DevOps, and AI adoption.
- **Strategic Enablement:** Designed and implemented processes, training, and governance to drive adoption of DevOps, cloud, and GenAI solutions across multiple teams and projects.
- **Business Impact & Transformation:** Proven ability to streamline software delivery, optimize cloud infrastructure, and enhance operational efficiency while ensuring compliance, reliability, and security.

SKILLS

- **DevOps & CI/CD:** Azure DevOps, Bitbucket, GitHub, Terraform, YAML Pipelines, and Kubernetes
- **Cloud & Containers:** Azure, Linode, Kubernetes (AKS) Docker, Helm
- **Security & DevSecOps:** SonarQube, CodeQL, Synk, S360, WCP, and threat modeling.

- **Monitoring & Observability:** Prometheus, Grafana, Azure Monitor, and Application Insights
- **Languages & Frameworks:** Python, Node.js, .NET Core, and Vue.js
- **Infrastructure as Code:** Terraform, Azure CLI, and Bash
- **iPaaS & Integration:** AFD, Application Gateway, and Azure API Management
- **Gen AI:** Familiar with Agentic AI solutioning using Lang Graph, AutoGen, and OpenAI SDK. Experienced on LLM Finetuning.

PROFESSIONAL EXPERIENCES

Microsoft | Dallas, TX | Jun 2025 –Current
Principal Architect, Automation, LTI Mindtree Ltd.
Project: GenAI Enablement Practice

GenAI Enablement Practice is an extension to the LTI Mindtree MSBU (Microsoft Business Unit) practice to explore, design, and operationalize generative AI solutions across software development, support, and DevOps. The team accelerates engineering productivity through AI-assisted development, automation, and intelligent tooling. It enables reliable, secure, and scalable adoption of GenAI by defining best practices, guardrails, and architectures. The team partners with product and engineering groups to identify high-impact use cases and deliver measurable value. It also drives knowledge sharing, experimentation, and continuous learning to embed GenAI into everyday workflows.

Role and Responsibilities

- **Helped shape and evolve the GenAI Enablement Practice** within the Microsoft Business Unit, contributing to early architecture patterns, tooling direction, and practical adoption of generative AI across Dev, Support, and DevOps teams.
- **Designed and guided the development of initial GenAI tools and prototypes**, including *SRE Buddy* (agentic chatbot using RAG over SOPs and integrations with ICM, Kusto DB, Azure DevOps, and external knowledge sources) to assist SREs in day-to-day operations.
- **Supported experimentation with fine-tuned LLM solutions** for incident classification, ticket routing, and cause-code analysis, helping teams improve operational insights and reduce manual triage effort.
- **Enabled data-driven reporting and analysis** through solutions like *AICA (AI Content Analysis)*, which analyzes bot-user interactions and generates management-ready insights and summaries.
- **Established a scalable hosting and delivery approach for GenAI workloads**, deploying services on Azure Kubernetes Service (AKS), enabling CI/CD via Azure DevOps pipelines, and introducing Langfuse and Grafana for continuous monitoring, observability, and prompt/response evaluation.
- **Encouraged adoption of Microsoft-native GenAI tooling**, promoting daily use of Azure SRE Agent and Microsoft MCP servers to improve engineering and operational productivity.
- **Contributed to the design of an internal MCP Hub**, providing visibility into MCP server adoption, usage patterns, and integration progress to support informed scaling and standardization decisions.
- **Facilitated GenAI knowledge enablement across teams** by coordinating and hosting training sessions with external vendors and subject-matter experts, helping teams build practical understanding of GenAI capabilities, limitations, and responsible usage.

Microsoft | Dallas, TX | Sep 2022 – Current
Principal Architect, DevOps, LTI Mindtree Ltd.
Project: MSGDCBADE

Microsoft GDCBADE is the Marketing Web team responsible for the development, operation, and support of more than 100 Microsoft Marketing websites. The LTI Mindtree team provides comprehensive DevOps and Site Reliability Engineering (SRE) support for these platforms.

The primary objective of this project is to deliver end-to-end infrastructure and DevOps support to ensure secure, reliable, and high-performing marketing websites. This includes:

- Infrastructure provisioning and management
- Design, implementation, and maintenance of CI/CD pipelines
- Ensuring application and code security
- Deployment of applications across environments
- Continuous monitoring of infrastructure and applications
- Ensuring high availability, reliability, and operational excellence

The project aims to provide a stable, scalable, and secure environment while delivering an optimal digital experience for Microsoft's end customers.

Role and Responsibilities

- Led the team delivering DevOps and SRE support for Microsoft's Global Development Center Web (GDC Web) team, which drives the development, deployment, and continuous support of Microsoft's Marketing Websites, ensuring high availability, reliability, and operational excellence.
- Migrated **40+ Microsoft.com blog & news websites to Azure Kubernetes Service (AKS)**, reducing infrastructure costs by **40%**.
- Built **self-healing, auto-scalable AKS clusters** using Terraform and Helm, ensuring high availability.
- Implemented **RBAC, pod security policies, image scanning, and Azure AD authentication** to secure Kubernetes workloads.
- Developed CI/CD pipelines and templates in **Azure DevOps**, integrating **SAST (CodeQL), Azure defender and automated testing**.
- Played an instrumental role in Secure SDLC implementation in the team by playing role of creating and reviewing threat model, privacy review, and BCDR documentation.
- Automated **infrastructure deployment with Terraform, Azure DevOps** reducing errors and improving efficiency.
- Automated SRE day-to-day operations with custom tools built on Azure using Azure Automation, Python, Langchain, and Azure Open AI.
- Worked on Microsoft SFI (**Secure Future Initiatives**), implementing **Zero Trust security models** including password-less security with **Azure AD and OAuth**.
- Built **centralized monitoring dashboards** using **Datadog, Prometheus, Grafana, and Azure Monitor**.

Tenet Healthcare | Dallas, TX | Feb 2022 – Sep 2022 Technical Architect, DevOps, NTT Data Services

The objective of this project was to modernize and standardize the application delivery platform for the Tenet Healthcare account by driving enterprise-wide adoption of Azure DevOps. The initiative focused on migrating legacy TFS projects, strengthening security and compliance, and establishing scalable CI/CD practices across diverse technology stacks. A key goal was to improve developer productivity, release reliability, and visibility while enabling a secure, cloud-native DevOps operating model.

Role and Responsibilities

- Spearhead the **Azure DevOps adaptation** within the **NTT Tenet Account**.

- **Successfully migrated 60+ projects from TFS to Azure DevOps**, streamlining version control (Git), work item tracking (Azure Boards), and reporting (Azure Monitor, Power BI).
- Designed and enforced security controls using **Azure AD, RBAC, Service Connections, and Pipeline Approvals**, ensuring compliance with best practices for access management, repository protection (branch policies)
- Integrated pipelines with SonarQube for code scanning to enforce Secure SDLC.
- Developed and optimized YAML-based CI/CD pipelines for applications built using **.NET, Node.js, Java, and Python, integrating with Kubernetes (AKS), and Helm to enhance automation**, deployment efficiency, and developer productivity.
- Collaborated with cross-functional teams to modernize development process, implement **Git flow branching strategy**, and containerized deployments using Docker and Kubernetes.
- Provided technical guidance and troubleshooting support, ensuring a seamless transition while minimizing downtime and operational risks.
- Conducted firsthand training and workshops to educate teams on Azure DevOps best practices, security automation, and cloud-native deployment strategies.

Tenet Healthcare | Dallas, TX | Nov 2021 – Feb 2022
Technical Architect, Software Development, NTT Data Services

The objective of this project was to modernize and consolidate Tenet Healthcare's consumer-facing websites by migrating them to a custom CMS built on ASP.Net Core, Vue.js, and Sitefinity, while improving performance, availability, and user experience. The initiative included implementing value-added services, integrating patient portal systems, and transitioning the infrastructure to Azure Kubernetes Service (AKS) and App Services.

Role and Responsibilities

- Led the **migration of forty-eight consumer websites** to a **custom CMS** built with **ASP.Net Core, Vue.js, and Sitefinity**, increasing site performance and availability to **99.99% and improvised the website Lighthouse score from single digits to 80%**.
- Implemented many values added services to the websites like Google Analytics, A/B testing, SEO improvement tools like Site Improve, and Dynamic Form Hooks.
- Integrated with patient portal services like EPX Payment, Physician Search, ER Status and Patient Training booking.
- Later migrated the sites from the legacy infrastructure to **Azure Kubernetes Service (AKS), Azure App Services, and Azure Functions**.
- Implemented DevOps practice using Azure DevOps and streamlined the code release process using CI/CD pipelines.
- Established **real-time monitoring and incident creation** using **Dynatrace / Keynote**.

Tenet Healthcare | Dallas, TX | Oct 2016 – Nov 2021
Manager, Application Support, Dell Services

The objective of this engagement was to provide stable, scalable, and cost-effective application support for Tenet Healthcare's enterprise platforms while meeting strict SLA and availability requirements. The focus was on improving incident response, reducing recurring issues through automation and root-cause analysis, and strengthening coordination across development, infrastructure, and vendors. A key goal was to enhance service reliability, optimize support operations, and transition applications to an efficient in-house support model.

Role and Responsibilities

- Managed a **\$5M/year enterprise tower portfolio** of Tenet healthcare applications that was providing application support to Tenet's **consumer websites, SharePoint portals, and BPM applications** ensuring high availability, reliability, and operational efficiency.
- Managed incident response and problem resolution, reducing downtime and improving service continuity through ITIL best practices using Service Now.
- Coordinated with development, infrastructure, third-party vendors, and service management streamlining deployments and post-release support.
- **Ensured compliance with SLAs (Service Level Agreements)** by tracking performance metrics and driving continuous improvement initiatives.
- Managed stakeholder communications, providing regular status updates and ensuring alignment with business priorities.
- Implemented initiative-taking monitoring and automation, enhancing system performance, and minimizing recurring issues.
- Directed support transitions for new applications, establishing knowledge bases, runbooks, and onboarding processes.
- Transitioned multiple applications from **third-party vendors to in-house support**, reducing operational costs.
- Reduced **ticket volume by 60% over two years** through process automation and cause code analysis.

Tenet Healthcare | Dallas, TX | May 2015 – Oct 2016 **Manager, Software Development, NTT Data Services**

This project was to consolidate and modernize Tenet Healthcare's SharePoint ecosystem, ensuring seamless access, data integrity, and improved user experience across hospital and consumer portals. The initiative focused on large-scale content and site migrations, multi-site transition management, and implementation of governance and access controls. Key goals included minimizing downtime, ensuring compliance, and enabling smooth adoption through stakeholder communication and end-user training.

Role and Responsibilities

- Led SharePoint Migration Project – Successfully managed the consolidation of legacy SharePoint sites into a centralized SharePoint 2013 Portal.
- Oversaw Large-Scale Data Migration – Managed the migration of 4TB of Document Management Sites, ensuring data integrity, compliance, and seamless access post-migration.
- Managed Multi-Site Transition – Directed the migration of 40+ Hospital Intranet Websites and 40+ Consumer Websites, optimizing performance, security, and user experience.
- End-to-End Project Ownership – Defined project scope, timelines, and deliverables while coordinating cross-functional teams, including IT, business stakeholders, and external vendors.
- Risk & Issue Management – Identified potential risks, developed mitigation strategies, and ensured minimal downtime and business disruption during migration.
- Stakeholder Communication – Functioned as the primary point of contact for executive leadership, providing status updates, progress reports, and addressing concerns proactively.
- Process Optimization & Governance – Established governance policies, access controls, and best practices for content management post-migration.

- Change Management & Training – Facilitated training sessions and provided user adoption support to ensure a smooth transition to the new SharePoint environment.

Dell Microsoft Practice | Bengaluru, India | Apr 2013 – Apr 2015 **Manager, Software Development, Dell Services**

This engagement was to establish and scale Dell's SharePoint Practice within the Microsoft horizontal, delivering standardized, industrialized solutions for SharePoint migration, knowledge management, and ITIL-based services. The initiative focused on driving delivery excellence through reusable assets, governance, and operational reviews, while building team capabilities via training, certification, and resource planning. Key goals included optimizing partner collaborations, ensuring consistent service quality, and managing a high-performing team across multiple locations.

Role and Responsibilities

- Managing SharePoint Practice within Microsoft horizontal in the Application Development and Maintenance Service line.
- Delivery Excellence – Institutionalize Reusable asset repository, coding standards, delivery operations governance, and operational reviews
- Industrialized Services – Develop fixed-price product offerings on SharePoint migration, Knowledge management, and ITIL. Associated sales collaterals and branding
- Capability Building – Plan for skill enhancement through external and internal training & certification programs, Resource demand-supply tracking, and staffing projections
- Partner engagement – Collaborate with Dell Software Group (former Quest) on building an optimized delivery model for cookie-cutter SharePoint migration offerings
- HR People management – Managing a team of twenty-five associates across Bengaluru and Noida

O & M | Bangalore, India | Nov 2011 – Mar 2013 **Technical Lead, Software Development, Dell Services**

The project was to modernize and streamline integration and monitoring processes for O&M applications by implementing a centralized orchestration layer using BizTalk 2010. The initiative focused on eliminating redundant point-to-point integrations, improving exception handling and failure monitoring through a centralized ESB portal, and enhancing alert management with a custom artifact monitoring tool. Key goals included increasing operational efficiency, reducing repeated failures, and providing transparent status reporting to stakeholders.

Role and Responsibilities

- Developed the **orchestration layer using BizTalk 2010**, replacing point-to-point integrations, and eliminating redundant middle-layer technologies.
- Built a **centralized repository for exception handling and failures** using **ESB (Enterprise Service Bus) Management Portal** for improved monitoring.
- Developed an **artifact monitoring tool** to control alert notifications and reduce repeated failure issues.
- Responsible for **WBS creation, stakeholder communication, and status reporting**.

Microsoft Amalga Implémentation | Bangalore, India | Sep 2009 – Nov. 2011 **Technical Analyst, Software Development, Dell Services**

This project was to implement and integrate the Microsoft Amalga UIS EMR across multiple hospital systems, ensuring seamless interoperability using HL7, XML, and CSV standards. The initiative focused on building and mentoring offshore Amalga competency teams, delivering successful implementations at major hospitals, and maintaining

compliance with Microsoft quality and audit standards. Key goals included improving EMR integration, operational efficiency, and enabling standardized healthcare data management across facilities.

Role and Responsibilities

- Worked **integration of hospital systems with Amalga UIS EMR**, using **HL7, XML, and CSV standards** for seamless interoperability.
- Built and trained **Amalga competency teams at offshore delivery centers** in India and China.
- Successfully implemented **Amalga UIS at major hospitals**, including **Moses Cone Hospital System (MCHS), Norton Healthcare, and St Joe Hospitals**.
- Ensured **quality deliverables, audits, and compliance with Microsoft standards**.

EDUCATION & CERTIFICATIONS

- Post Graduate Diploma in IT, VGSOM, IIT Kharagpur, India
- Bachelor of Technology, Electrical Engineering, CET, OUAT Bhubaneswar, India
- AZ-400: Microsoft Certified DevOps Engineer Expert
- AZ-104: Microsoft Certified Azure Administrator
- CKAD: Certified Kubernetes Application Developer
- PSM I: Certified Professional Scrum Master
- Terraform Associate Certification
- HL7 Healthcare Messaging Certification
- Post Graduate in AI/ML and GenAI – In Progress



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